

JEE MAIN REVISION MASTER: LINEAR PROGRAMMING 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Linear Programming — Standard Calculations & Problem Archetypes	11 PYQs	Quantitative Analysis
Moderate	Linear Programming — Core Principles & Governing Laws	3 PYQs	Core Principle
Moderate	Linear Programming — Applied Cases, Reactions & Traps	3 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Linear Programming — Standard Calculations & Problem Archetypes	$I_1 \rightarrow \frac{x}{4} + \frac{y}{5} = 1, \quad x=0, \quad z=11x+7y$
Linear Programming — Core Principles & Governing Laws	$Z = ax + by, \quad \max z = c_1x_1 + c_2x_2 + \dots + c_nx_n, \quad a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \leq b_i,$
Linear Programming — Applied Cases, Reactions & Traps	$x+y=7, \quad \begin{aligned} 2x-3y+6=0 \\ x,y \geq 0 \end{aligned} \text{ (ii)}, \quad z=x+4y$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Linear Programming — Standard Calculations & Problem Archetypes (High Yield — 11 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Linear Programming.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The corner points of the feasible region determined by the system of linear constraints are $(0,10), (5,5), (15,15), (0,20)$. Let $z=px+qy$ wh..." [Ans: D]
- [2025 Shift] Q#4 (MCQ): "The maximum value of $z=3x+4y$, subject to the constraints $x+y \leq 40, x+2y \leq 60$ and $x,y \geq 0$ is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "Corner points of the feasible region for an LPP are $(0,2), (3,0), (6,0), (6,8)$ and $(0,5)$. Let $Z=4x+6y$ be the objective function. The min..." [Ans: N/A]

3.2 Linear Programming — Core Principles & Governing Laws (Moderate Yield — 3 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Linear Programming.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Linear Programming.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Linear Programming PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points ..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Consider the following statements: Statement (I): In a LPP, the objective function is always linear. Statement (II): In a LPP, the linear ine..." [Ans: N/A]
- [2017 Shift] Q#3 (MCQ): "If an LPP admits optimal solution at two consecutive vertices of a feasible region, then..." [Ans: N/A]

3.3 Linear Programming — Applied Cases, Reactions & Traps (Moderate Yield — 3 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Linear Programming.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "The shaded region in the figure given is the solution of which of the inequations?..." [Ans: D]
- [2020 Shift] Q#2 (MCQ): "Corner points of the feasible region determined by the system of linear constraints are $(0,3), (1,1)$ and $(3,0)$. Let $z=px+qy$, where, $p, q > 0$..." [Ans: N/A]
- [2018 Shift] Q#2 (MCQ): "For the LPP, maximize $z=x+4y$ subject to the constraints $x+2y \leq 2, x+2y \geq 8, x,y \geq 0$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points at which $Z_{\text{(max)}}$ occurs is

- (A) 1
- (B) 2
- (C) 0
- (D) Infinity

Correct Answer: None

Solution Analysis: If the objective function $Z=ax+by$ has the same maximum value at two corner points of the feasible region, then Z will have the same value at every point on the line segment joining those two corner points. This happens because $Z=ax+by$ is a linear function. So, if it is maximum at two endpoints of an edge of the feasible region, then it remains maximum at all points on that edge. Hence, $Z_{\text{(max)}}$ occurs at infinitely many points. So, the correct option is: Option D: Infinity

Benchmark Problem 2: KCET 2026 (Q#2)

The corner points of the feasible region determined by the system of linear constraints are $(0,10), (5,5), (15,15), (0,20)$. Let $z=px+qy$ where $p,q>0$. The relation between p and q , so that the maximum z occurs at both points $(15,15)$ and $(0,20)$ is

- (A) $p=q$
- (B) $p=2q$
- (C) $q=2p$
- (D) $q=3p$ [Correct]

Correct Answer: D

Solution Analysis: For the maximum value of $z=px+qy$ to occur at both points $(15,15)$ and $(0,20)$, the value of z at these two points must be equal. At $(15,15)$, $z=15p+15q$ At $(0,20)$, $z=0p+20q=20q$ Since maximum occurs at both points, $15p+15q=20q$ $15p=5q$ $q=3p$ So, the required relation is $q=3p$ Hence, the correct option is Option D.

Benchmark Problem 3: KCET 2023 (Q#2)

The shaded region in the figure given is the solution of which of the inequations?

- (A) $x+y \geq 7, 2x-3y+6 \leq 0, x \geq 0, y \geq 0$
- (B) $x+y \geq 7, 2x-3y+6 \geq 0, x \geq 0, y \geq 0$
- (C) $x+y \leq 7, 2x-3y+6 \leq 0, x \geq 0, y \geq 0$
- (D) $x+y \leq 7, 2x-3y+6 \geq 0, x \geq 0, y \geq 0$ [Correct]

Correct Answer: D

Solution Analysis: The line joining $A(7,0)$ and $(0,7)$ is $x+y=7$ (i) And the line joining $C(0,2)$ and $B(3,4)$ is $2x-3y+6=0$ $x,y \geq 0$ (ii) The shaded region is bounded by $x+y \leq 7, 2x-3y+6 \geq 0, x \geq 0, y \geq 0$ Hence, option (d) is correct option.